

Various open source blockchain products and their critical characteristics

Feature	Open source product	Remarks
Public blockchain	Ethereum, Ripple, Litecoin, etc.	Permissionless blockchain, participants are anonymous and anyone can participate. There is no centralised controlling authority.
Private blockchain	Hyperledger projects, Exonum, OpenChain, MultiChain	Permissioned blockchain, identity of the participants is known before, no need of mining and native cryptocurrency to pay for mining.
Federated blockchain (Consortium)	R3, EWF, etc.	Group of companies or individuals coming together to make decisions for the benefit of the whole network. Such groups are also called consortiums or a federated blockchain.
Smart contracts in general programming languages	Hyperledger, Ethereum	Ethereum will not directly support Java but can do so, with the help of a framework like web3j.
No mining and native cryptocurrency	Hyperledger, Corda, Openchain	Typically, public blockchains require mining (as part of proof of work) to add a block to the chain. Miners will be rewarded by using native cryptocurrency.
Pluggable consensus algorithms	Hyperledger, Corda, Openchain	Support for pluggable consensus used for different applications.
Pluggable ordering service	Hyperledger	Support for pluggable ordering service used for different applications.
Pluggable membership service	Hyperledger	Support for pluggable membership used for onboarding new users onto the network.

- Small organisations (with no business network)
- Those looking for a database replacement
- Those looking for a messaging solution
- Those in search of transaction processing replacement

Blockchain has shown its potential for transforming traditional industries with its key characteristics — decentralisation, persistence, anonymity and auditability. Blockchain based enterprise applications increase the effectiveness of an enterprise, reduce cost of transactions and speed up interactions between the enterprises and its customers.

Blockchain provides better security during transactions of any value. It is a unique and a universal technology that helps to streamline and automate nearly all customer services or legal contracts, while increasing the transparency and effectiveness of enterprises. However, a lot of exploration is needed today in domains applying blockchain technology across various business units — on how to minimise enterprise costs, improve security in an era of cyber uncertainty and enhance customer delivery. **END** 

Acknowledgements

The authors would like to thank Hari Kishan Burle and Raju Alluri of the Global Enterprise Architecture Group of Wipro Technologies for their time and support to bring this article out as part of the Architecture Practice effort.

Disclaimer

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